MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3410 PRESS. OXDN. AUTOCL. FEED STORAGE
DESIGN OF TANK FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
PRESS	:	Pressure
OXDN	:	Oxidation
AUTOCL	:	Autoclave

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1. INTENT

This document is to prepare design of foundation for **POX FEED TANK & POX RECYCLE TANK (3410-REA-001,002,003 & 004)**. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel. For member strength calculation force values are directly taken from Equipment load data drawings & designed in excel sheet.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Load Data drawings attached in Attachment 1.Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2 was also referred.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

5.3 Foundations

Foundation shall be Earth Foundation with a Concrete Ring wall for Tanks. The design calculations are prepared as per recommendation of standard API 650 - Annex B.

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5.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation	75
Exposed to weather or contact with ground	Wall	50

5.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 350 Kn/m^2 at 1.25 m below EGL for the tank foundations. (Refer Table 5 BH-9 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).

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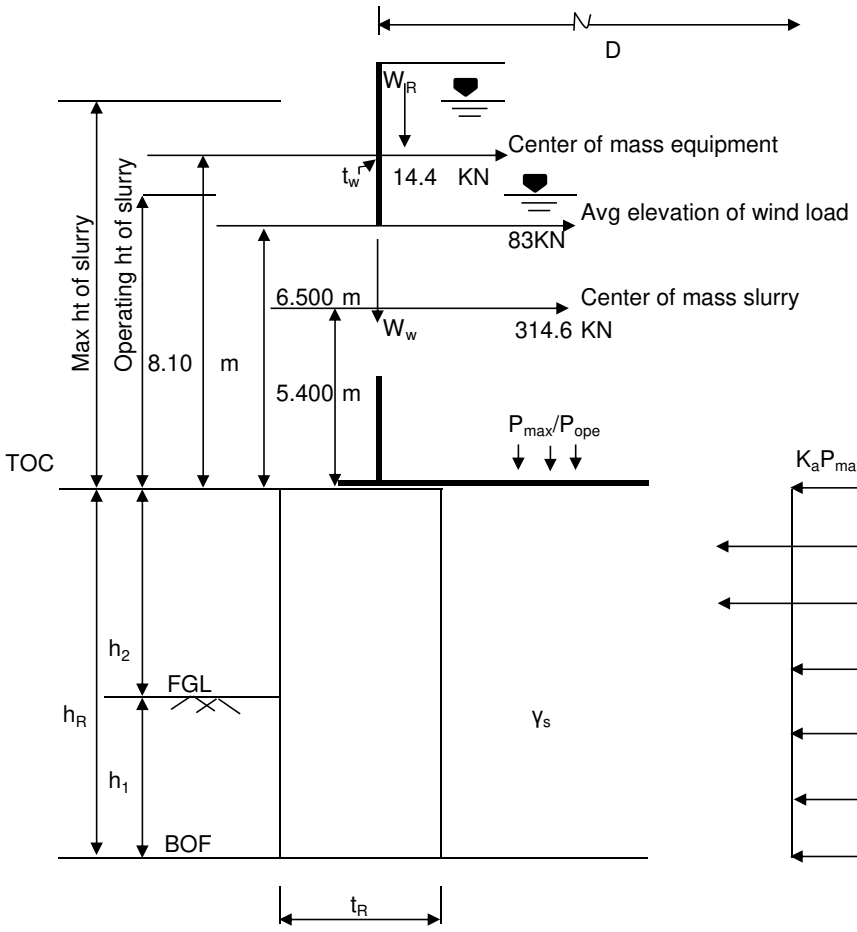
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6. DESIGN CALCULATION FOR FOUNDATION

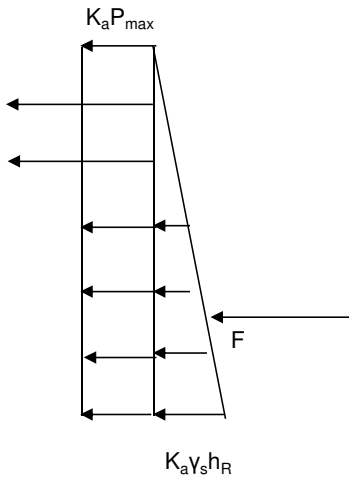
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Design Calculation of Foundation for POX FEED TANK (3410-REA-001 & 002)



Typical Section Through Tank



Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	8.0	m
Self weight of tank =	327.50	KN
Self weight of tank per m of circumference,W_w =	13.03	KN/m
External platform weight =	36.5	KN
Self weight of tank roof per m of circumference,W_R=	1.452	KN/m
Maximum weight of slurry=	8055	KN
Pressure at base of tank ,P_{max}	160.2	KN/m^2
Operating weight of slurry=	7160	KN
Pressure at base of tank ,P_{ope}	142.4	KN/m^2

Foundation Material :

Concrete, f_c' =	35	Mpa	
Reinforcement, f_y =	420	Mpa	
Unit weight of concrete,γ_c =	24	Kn/m^3	

Soil Data :

Unit weight of soil,γ_s =	19	Kn/m^3	(Natural sand)
Allowable bearing pressure, q_{all}=	350	Kn/m^2	
Coefficient of active earth pressure,K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	2.4	m
FGL to Bottom of foundation ,h_1 =	1.25	m
FGL to Top of concrete ,h_2 =	1.15	m
Ringwall thickness, t_R =	450	mm
Inner dia of ringwall,D_i =	7.55	m
Outer dia of ringwall,D_o =	8.45	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	21.48	m^3

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Check of soil bearing pressure (Max Slurry Load): For wind Load

Wind force = 83 kn acting at a height of 8.900 m from BOF

M_{OT} = Overturning moment = 738.7 KN-m

M_{OT} / S_{xx} = 34.4 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 170 KN/m^2

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 204 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 136 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = (W_w + W_R)/t_R + h_R\gamma_c$ = 89.8 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 124.2 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 55.40 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 738.7 KN-m

M_{OR} = Restoring moment = (327.5+PI()*8*0.45*2.4*25)*8/2 = 4024.3 KN-m

FOS = M_{OR} / M_{OT} 5.45 > 1.5 ok

Check for soil bearing pressure (Operating Slurry Load): For Seismic Load

Seismic force at center of mass equipment = 327.5/(327.5+7160)*329 = 14.39 KN
acting at a height of 10.500 m from BOF

Seismic force at center of mass slurry = 7160/(327.5+7160)*329 = 314.6 KN
acting at a height of 7.800 m from BOF

M_{OT} = Overturning moment = 2605 KN-m

M_{OT} / S_{xx} = 121 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{ope}/2 + h_R\gamma_c$ = 161 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 246 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 76.1 KN/m^2

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = (W_w + W_R)/t_R + 0.7 \times P_{max} + h_R\gamma_c =$ 202.0 Kn/m^2 < 350.0 KN/m^2 Ok

As the tank encroaches the ring wall by 300mm out of total 450mm , the pressure under the tank is reduced to 300/450 = 0.7 times.

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s =$ 205.8 Kn/m^2 < 350.0 KN/m^2 Ok

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2) =$ 145 Kn/m

Hoop tension in ringwall, $T = 1/2FD =$ 580 Kn

Factored design tension, $T_u =$ 812 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y =$ 2148 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R =$ 2160 mm^2

$A_{s,reqd}$ (per face of ring wall)= 1080 mm^2

Required : 12 dia bar @ 251.5 c/c (per face)

Provide : 12 dia bar @ 175 c/c (per face)

$A_{s,\ provided}$(per face) = 1552 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R =$ 540 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 270 mm^2

Required : 10 dia bar @ 291.1 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

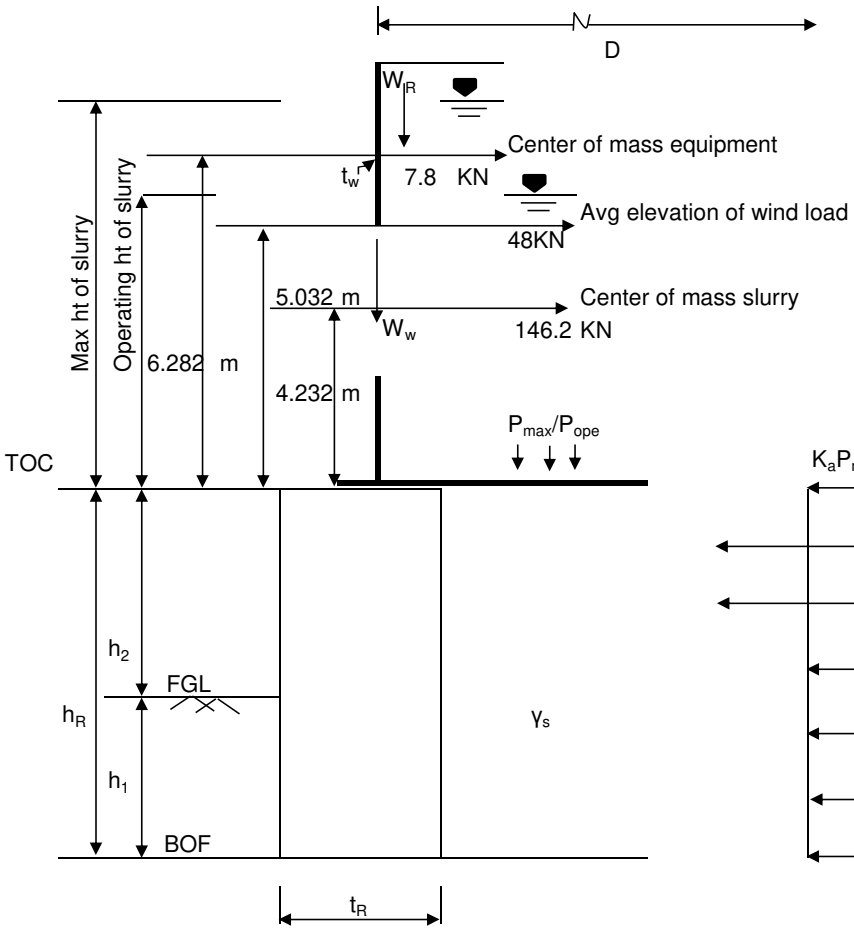
$A_{s,\ provided}$(per face) = 392.9 mm^2 ok

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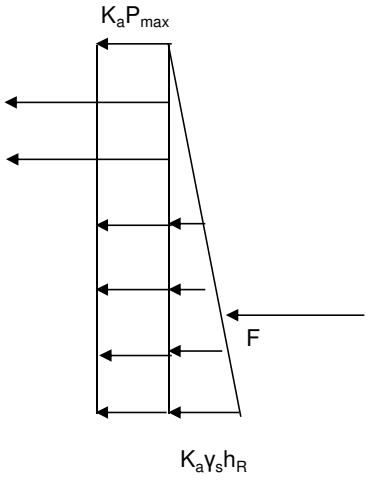
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Design Calculation of Foundation for POX RECYCLE TANK (3410-REA-003 & 004)



Typical Section Through Tank



Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	5.5	m
Self weight of tank =	183.00	KN
Self weight of tank per m of circumference,W_w =	10.59	KN/m
External platform weight =	24	KN
Self weight of tank roof per m of circumference,W_R=	1.389	KN/m
Maximum weight of slurry=	3748	KN
Pressure at base of tank ,P_{max}	157.8	KN/m^2
Operating weight of slurry=	3417	KN
Pressure at base of tank ,P_{ope}	143.8	KN/m^2
Seismic moment at base level =	668	KN-M

Foundation Material :

Concrete, f_c' =	35	Mpa	
Reinforcement, f_y =	420	Mpa	
Unit weight of concrete,γ_c =	24	Kn/m^3	

Soil Data :

Unit weight of soil,γ_s =	19	Kn/m^3	(Natural sand)
Allowable bearing pressure, q_{all}=	350	Kn/m^2	
Coefficient of active earth pressure,K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	2.25	m
FGL to Bottom of foundation ,h_1 =	1.25	m
FGL to Top of concrete ,h_2 =	1.0	m
Ringwall thickness, t_R =	450	mm
Inner dia of ringwall,D_i =	5.05	m
Outer dia of ringwall,D_o =	5.95	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	9.95	m^3

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Check of soil bearing pressure (Max Slurry Load): For wind Load

Wind force = 48 kn acting at a height of 7.282 m from BOF

M_{OT} = Overturning moment = 349.5 KN-m

M_{OT} / S_{xx} = 35.1 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 159 KN/m^2

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 195 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 124 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = (W_w + W_R)/t_R + h_R\gamma_c$ = 80.6 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 115.8 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 45.49 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 349.5 KN-m

M_{OR} = Restoring moment = (183+PI()*5.5*0.45*2.25*25)*5.5/2 = 1706.0 KN-m

FOS = M_{OR} / M_{OT} 4.88 > 1.5 ok

Check for soil bearing pressure (Operating Slurry Load): For Seismic Load

Seismic force at center of mass equipment = {668/(3417*4.232+183*6.282)}*183 = 7.831 KN

acting at a height of 8.532 m from BOF

Seismic force at center of mass slurry = {668/(3417*4.232+183*6.282)}*3417 = 146.2 KN

acting at a height of 6.482 m from BOF

M_{OT} = Overturning moment = 1015 KN-m

M_{OT} / S_{xx} = 102 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{ope}/2 + h_R\gamma_c$ = 153 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 224 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 81.1 KN/m^2

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = (W_w + W_R)/t_R + 0.76 \times P_{max} + h_R\gamma_c =$ 200.5 Kn/m^2 < 350.0 KN/m^2 Ok

As the tank encroaches the ring wall by 340mm out of total 450mm , the pressure under the tank is reduced to 340/450 = 0.76 times.

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s =$ 200.5 Kn/m^2 < 350.0 KN/m^2 Ok

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2) =$ 133 Kn/m

Hoop tension in ringwall, $T = 1/2FD =$ 366 Kn

Factored design tension, $T_u =$ 512 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y =$ 1355 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R =$ 2025 mm^2

$A_{s,reqd}$ (per face of ring wall)= 1013 mm^2

Required : 12 dia bar @ 251.5 c/c (per face)

Provide : 12 dia bar @ 175 c/c (per face)

$A_{s,\ provided}$(per face) = 1455 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R =$ 540 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 270 mm^2

Required : 10 dia bar @ 291.1 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

$A_{s,\ provided}$(per face) = 392.9 mm^2 ok

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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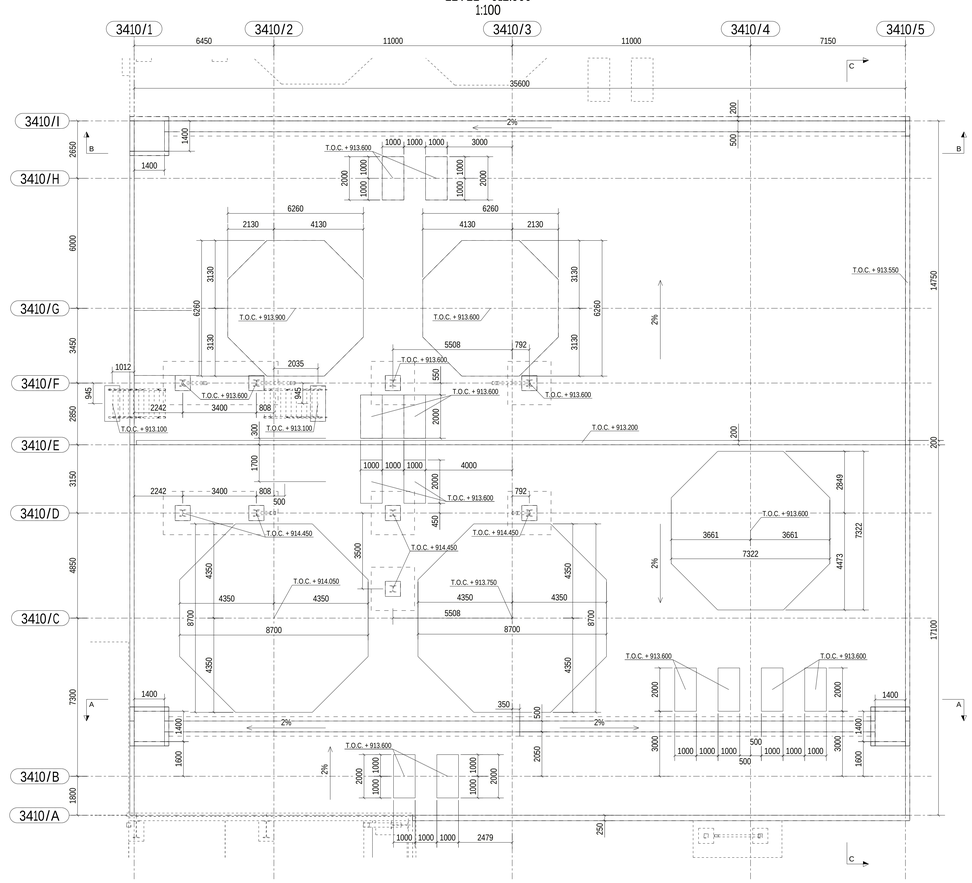
ATTACHMENT 1: LOAD DATA DRAWINGS

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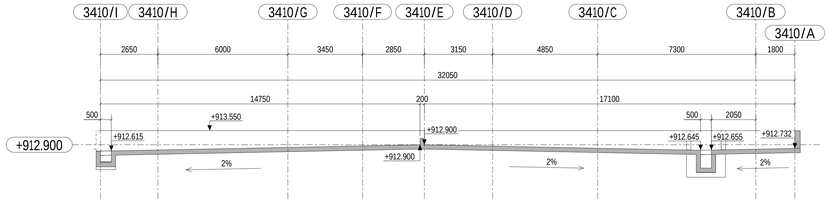
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DRAWING NUMBER MD-216-3410-EG-CV-GAD-1001 REV A01

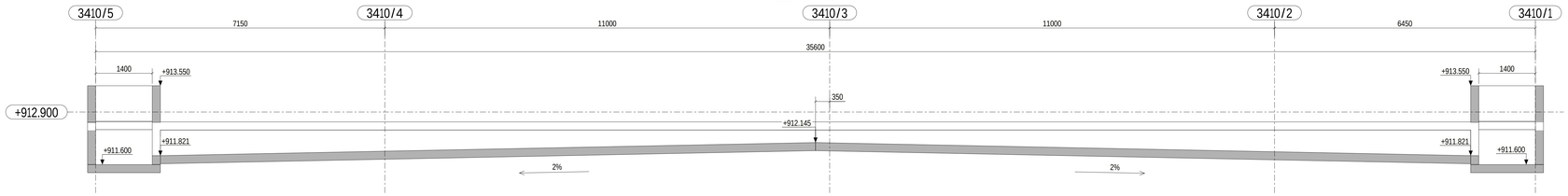
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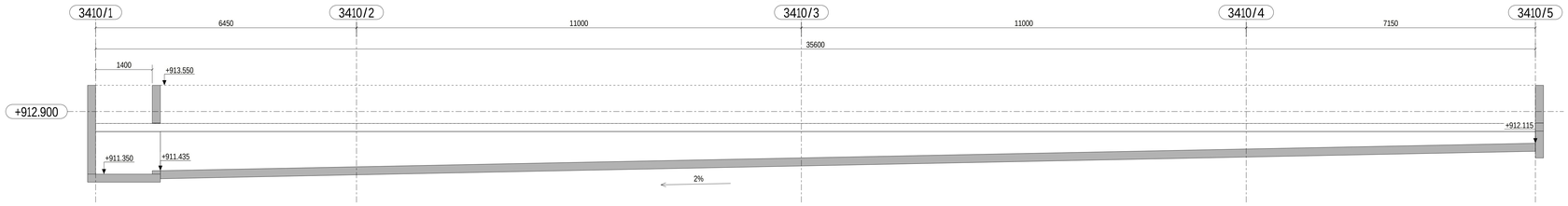
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A - A
1:50



B - B
1:50



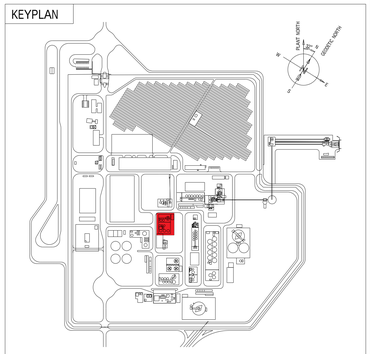
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3410 = RL(+)912.900 m

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ACID RESISTANT COATING
- CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



Document Number
MD-216-3410-EG-CV-GAD-1001

Revision A01

Approved

Mr Doron Golan - Bechtel Global

Jan 17, 2020, 10:08 PM GMT+2:00

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DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3410-EG-ST-GAD-1001	PLAN AT EL. +912.900 - GRID C-G/1-5										
MD-216-3420-EG-ME-GAD-1002	PLAN AT EL. +912.900 TO EL. +930.930	MD-216-3410-EG-CV-GAD-1002	PLAN AT EL. +912.900 - GRID I-M/1-5								
MD-216-1000-EG-ST-ORT-0001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3420-EG-ME-GAD-1001	PLAN AT EL. +912.900								

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	20.12.2019	ISSUED FOR APPROVAL	NikJan	MikLai	LuiPiz	SouCha	

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OUTOTEC DRAWING NUMBER: 900707_GAU01_HH12_CTA04_34101 — DOCUMENT ID: OU100667780 — REVISION: 0

DRAWN	NikJan	20.12.2019
CHECKED	MikLai	20.12.2019
ENGINEER	LuiPiz	20.12.2019
APPROVED	SouCha	20.12.2019
EMPLOYER		
SCALE	1:100	A0

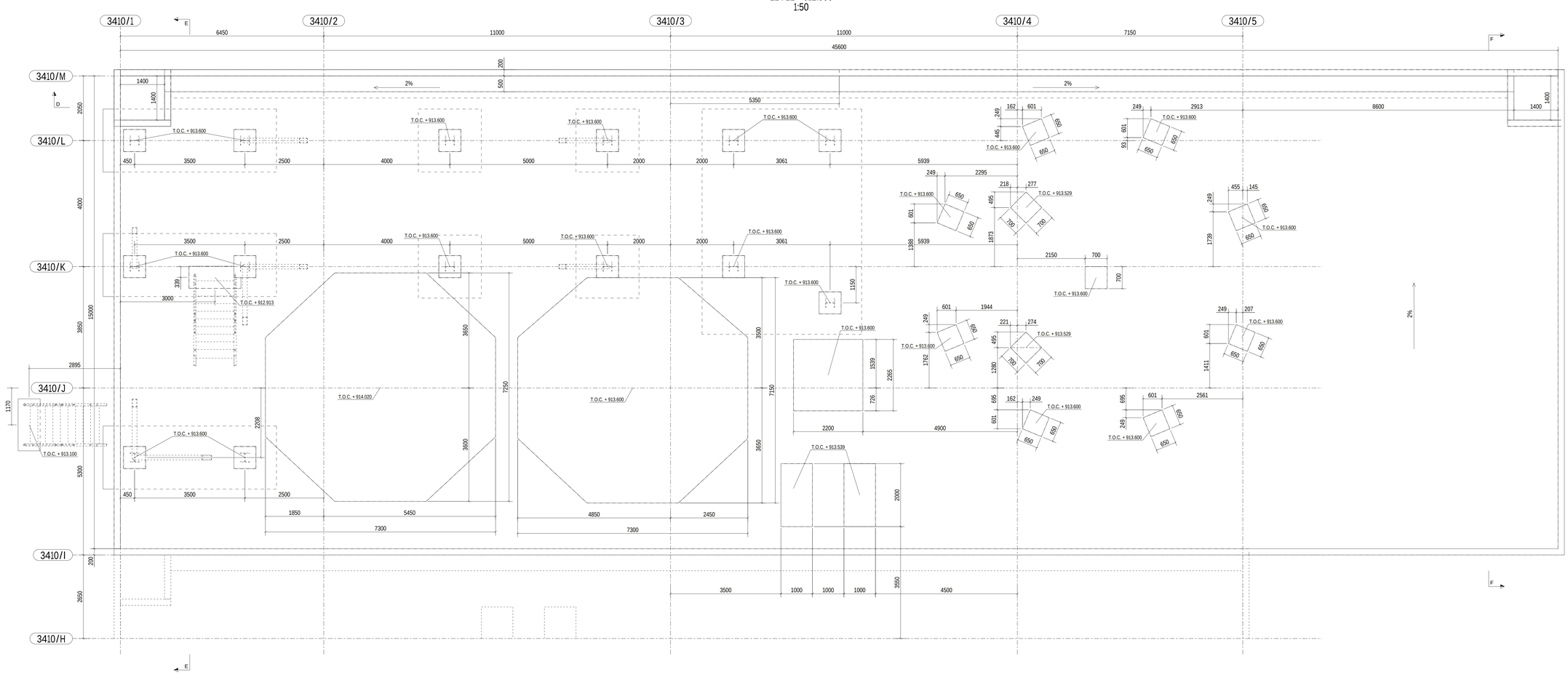
MANSOURAH - MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
PRESSURE OXIDATION - AUTOCLAVE FEED STORAGE
PLAN AT EL. +912.900 - GRID A-I/1-5

DRAWING NUMBER: MD-216-3410-EG-CV-GAD-1001 — REV: A01

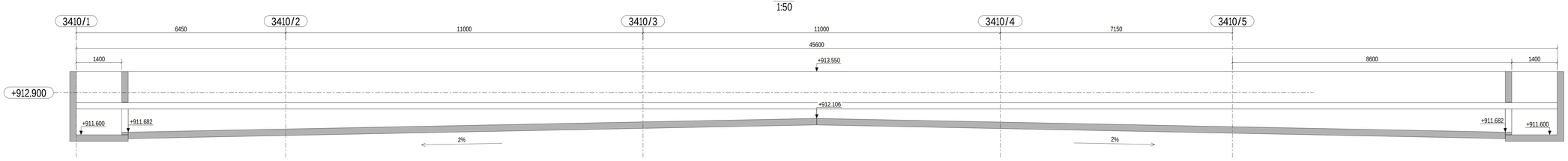
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DRAWING NUMBER: MD-216-3410-EG-CV-GAD-1002 | REV: A01

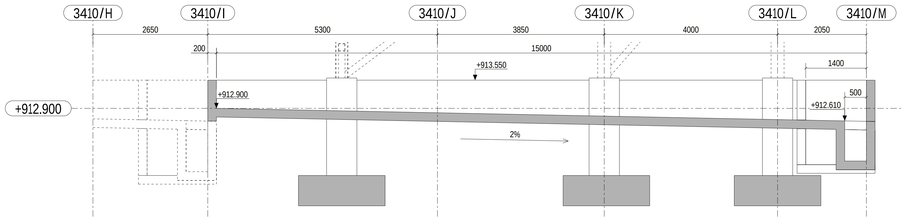
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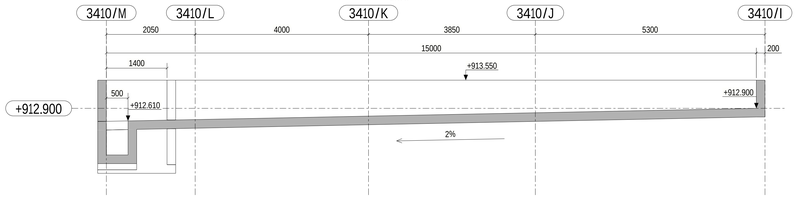
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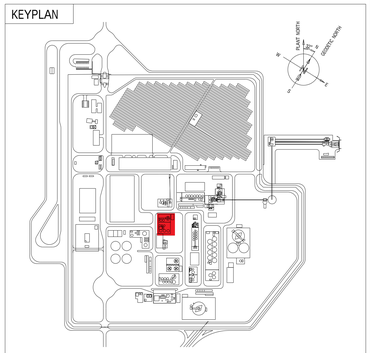
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3410 = RL(+)912.900 m

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ACID RESISTANT COATING
- CONCRETE WALL

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5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
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KEYPLAN



Document Number
MD-216-3410-EG-CV-GAD-1002

Revision A01

Approved

Mr Doron Golan - Bechtel Global

Jan 17, 2020, 10:09 PM GMT+2:00

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DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3410-EG-ST-GAD-1002	PLAN AT EL. +912.900 ~ GRID I-M/1-5
MD-216-3420-EG-ME-GAD-1002	PLAN AT EL. +912.900 TO EL. +930.930
MD-216-1000-EG-ST-CRT-0001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN BASIS

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3410-EG-CV-GAD-1001	PLAN AT EL. +912.900 ~ GRID A-I/1-5
MD-216-3420-EG-ME-GAD-1001	PLAN AT EL. +912.900

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	20.12.2019	ISSUED FOR APPROVAL	NikJan	MikLai	LuiPiz	SouCha	EMP

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_C_G_24930_001 | REVISION: 0

OUTOTEC DRAWING NUMBER: 900707_GAU01_HH12_CTA04_34102 | DOCUMENT ID: OU100667781 | REVISION: 0

DRAWN	NikJan	20.12.2019
CHECKED	MikLai	20.12.2019
ENGINEER	LuiPiz	20.12.2019
APPROVED	SouCha	20.12.2019
EMPLOYER		
SCALE	1:50	A0

MANSOURAH - MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
PRESSURE OXIDATION - AUTOCLAVE FEED STORAGE
PLAN AT EL. +912.900 - GRID I-M/1-5

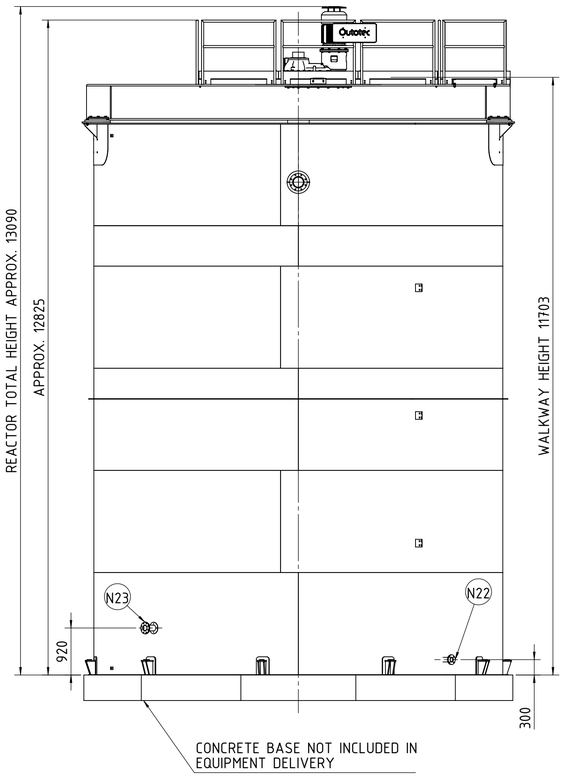
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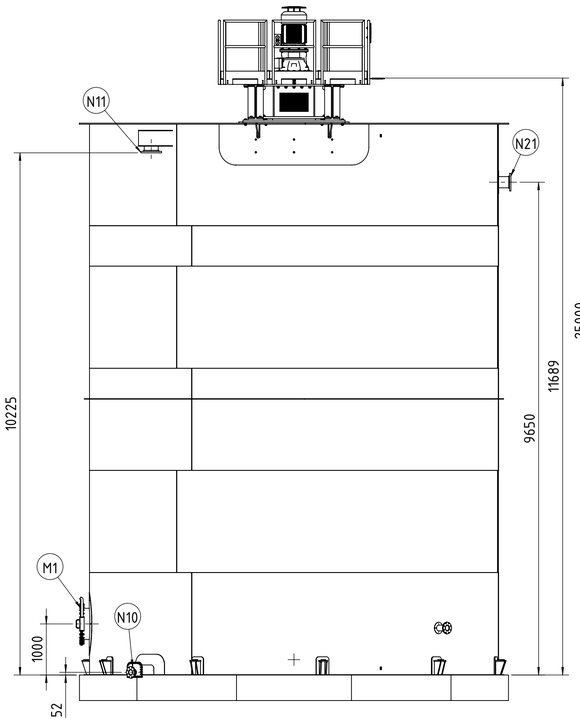
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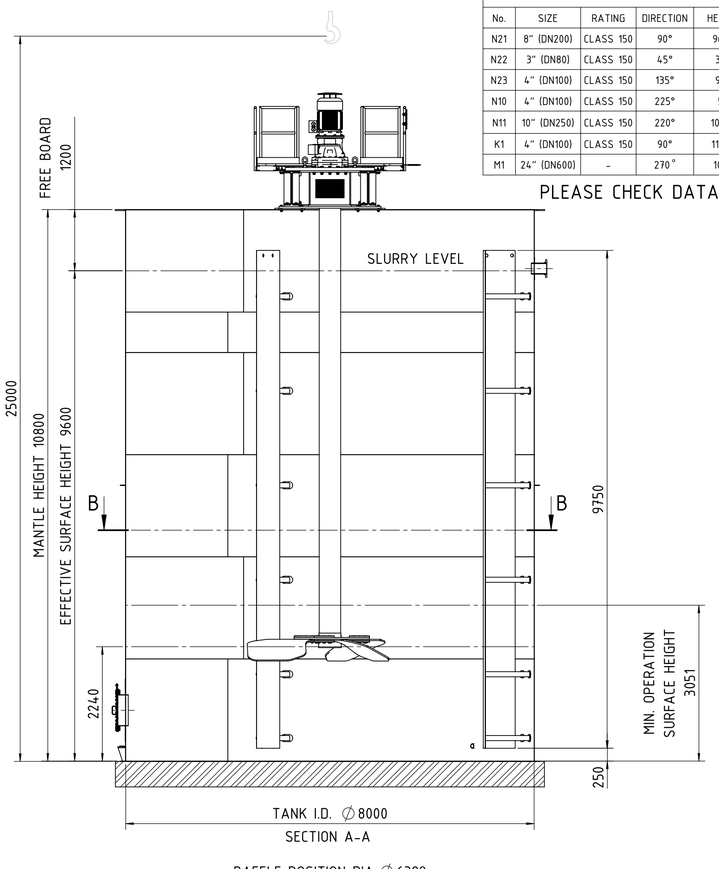
DRAWING NUMBER MD-216-3410-EG-ME-GAD-1007 REV A01

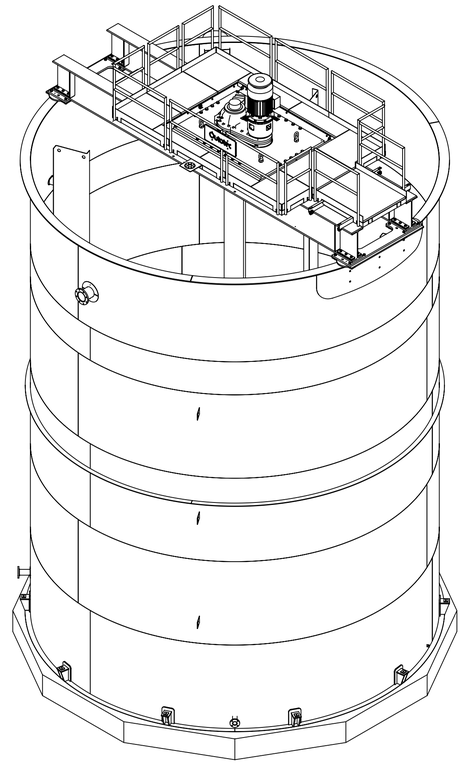
NOZZLE TABLE								
No.	SIZE	RATING	DIRECTION	HEIGHT	STANDARD	SERVICE	QUANTITY	REMARKS
N21	8" (DN200)	CLASS 150	90°	9650	ASME B16.5	SLURRY OUTLET	1	
N22	3" (DN80)	CLASS 150	45°	300	ASME B16.5	DENSITY MEASUREMENT PUMP	1	
N23	4" (DN100)	CLASS 150	135°	920	ASME B16.5	SLURRY OUTLET	1	
N10	4" (DN100)	CLASS 150	225°	52	ASME B16.5	DRAIN	1	
N11	10" (DN250)	CLASS 150	220°	10225	ASME B16.5	OVERFLOW	1	
K1	4" (DN100)	CLASS 150	90°	11689	ASME B16.5	LEVEL MEASUREMENT	1	
M1	24" (DN600)	-	270°	1000	OUTOTEC	MANHOLE	1	

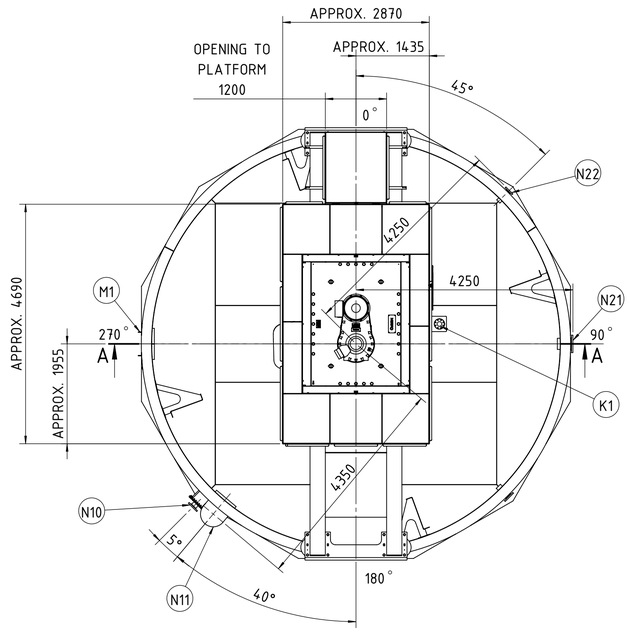
PLEASE CHECK DATASHEET MD-216-3410-EG-ME-DAT-1001 FOR PROCESS VALUES

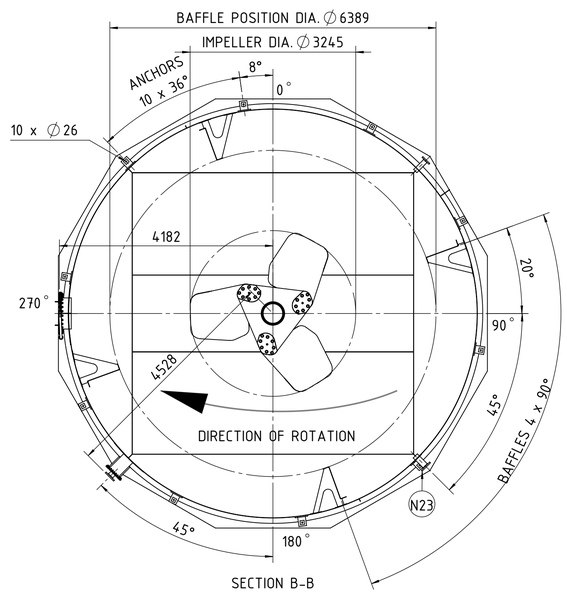












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Document Number
MD-216-3410-EG-ME-GAD-1007

Revision A01

Approved

Mr Doron Golan - Bechtel Global
Jan 17, 2020, 10:21 PM GMT+2:00

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POX FEED TANK	
DEAD WEIGHT (kg)	32800
OPERATIONAL WEIGHT (kg)	748800
MAXIMUM WEIGHT (kg)	838300

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3410-EG-ME-PLN-1001	FOUNDATION AND LOAD PLAN								
MD-216-3410-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3410-EG-PR-PID-1002	P&I DIAGRAM								
MD-216-3410-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	19.12.2019	ISSUED FOR APPROVAL	AT	TM	JR	GR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER 019045-450-M-G-02647-001 REVISION 0

OUTOTEC DRAWING NUMBER 900707 GAU01 HH12 MTL03 34101 DOCUMENT ID OU601239079 REVISION 0

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DRAWN	aritik	18.12.2019
CHECKED	tuoman	19.12.2019
ENGINEER	juhris	19.12.2019
APPROVED	GeoRie	19.12.2019
EMPLOYER		
SCALE	1:60	A1

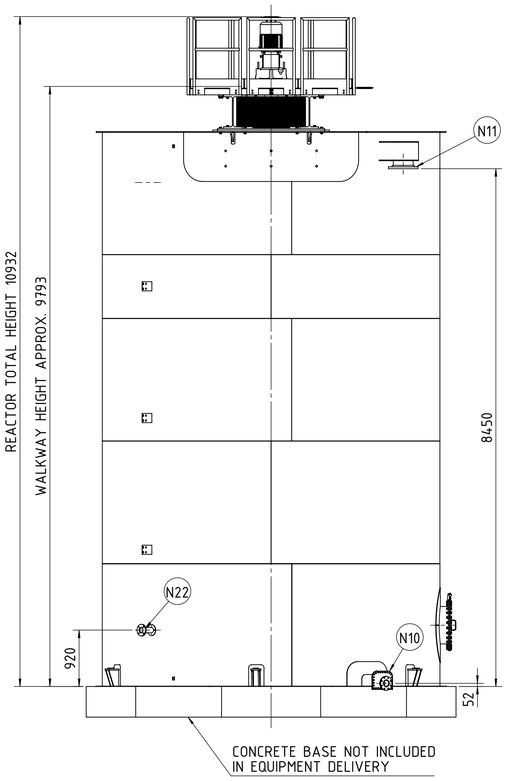
MANSOURAH-MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
PRESSURE OXIDATION
3410-REA-001 POX FEED TANK

DRAWING NUMBER MD-216-3410-EG-ME-GAD-1007 REV A01

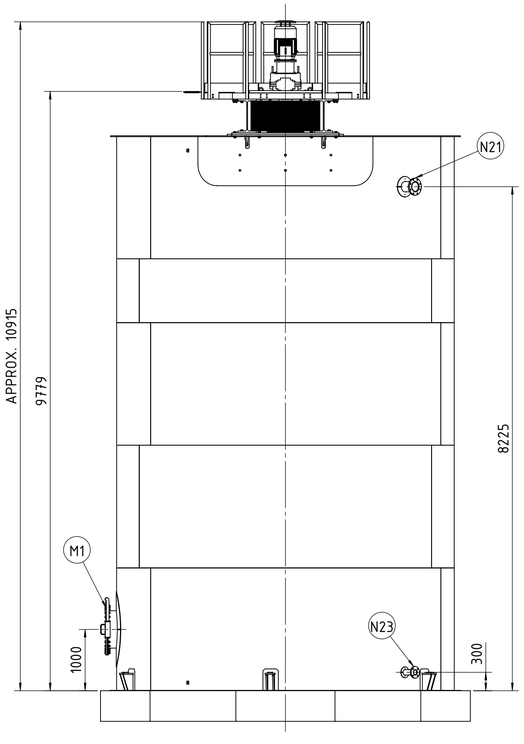
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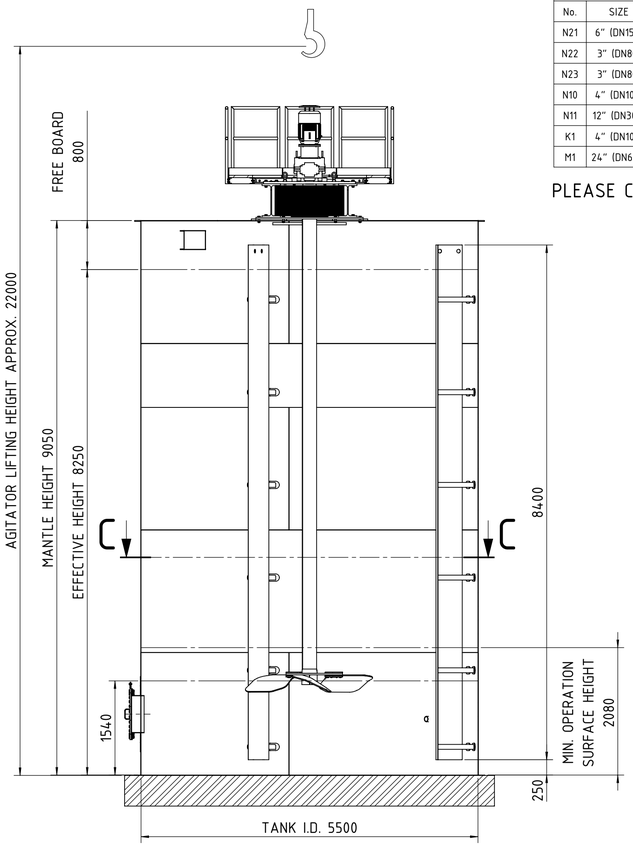
NOZZLE TABLE								
No.	SIZE	RATING	DIRECTION	HEIGHT	STANDARD	SERVICE	QUANTITY	REMARKS
N21	6" (DN150)	CLASS 150	135°	8225	ASME B16.5	SLURRY OUTLET	1	
N22	3" (DN80)	CLASS 150	45°	920	ASME B16.5	PUMP OUTLET	1	
N23	3" (DN80)	CLASS 150	135°	300	ASME B16.5	DENSITY PUMP	1	
N10	4" (DN100)	CLASS 150	325°	52	ASME B16.5	DRAIN	1	
N11	12" (DN300)	CLASS 150	315°	8450	ASME B16.5	OVERFLOW	1	
K1	4" (DN100)	CLASS 150	270°	9779	ASME B16.5	LEVEL MEASUREMENT	1	
M1	24" (DN600)	-	270°	1000	OUTOTEC	MANHOLE	1	

PLEASE CHECK DATASHEET MD-216-3410-EG-ME-DAT-1002 FOR PROCESS VALUES

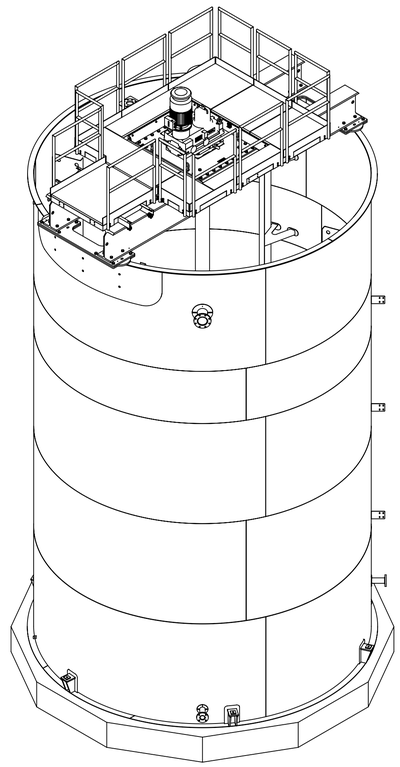


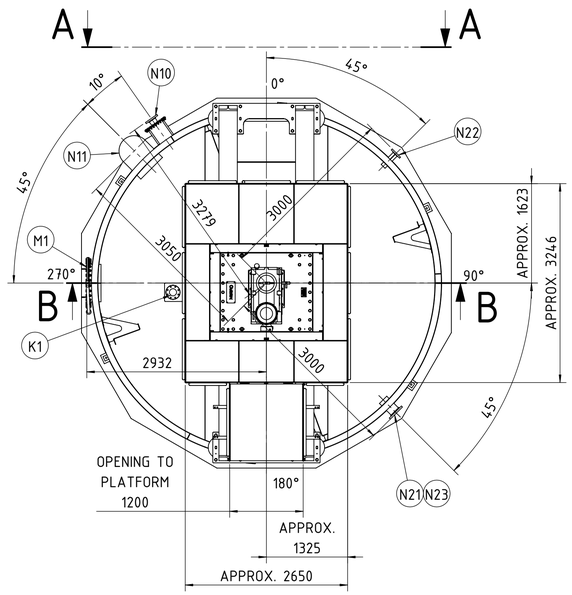
SECTION A-A

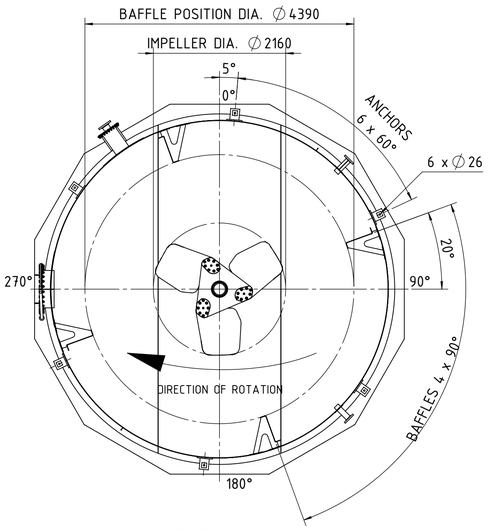




SECTION B-B







SECTION C-C

POX RECYCLE TANK	
DEAD WEIGHT (kg)	18300
OPERATIONAL WEIGHT (kg)	359900
MAXIMUM WEIGHT (kg)	393000

BECHTEL

Document Number
MD-216-3410-EG-ME-GAD-1008

Revision A01

Approved

Mr Doron Golan - Bechtel Global
Jan 17, 2020, 10:22 PM GMT+2:00

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DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3410-EG-ME-PLN-1003	FOUNDATION AND LOAD PLAN								
MD-216-3410-EG-ME-DAT-1002	EQUIPMENT DATASHEET								
MD-216-3410-EG-PR-PID-1012	P&I DIAGRAM								
MD-216-3410-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	19.12.2019	ISSUED FOR APPROVAL	AT	TM	JR	GR	

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LARSEN & TOUBRO DRAWING NUMBER 019045-450-M-G-02647-003 REVISION 0

OUTOTEC DRAWING NUMBER 900707 GAU01 HH12 MTL03 34103 DOCUMENT ID 0U601239427 REVISION 0

DRAWN	aritik	20.11.2019
CHECKED	tuoman	19.12.2019
ENGINEER	juhris	19.12.2019
APPROVED	GeoRie	19.12.2019
EMPLOYER		
SCALE	1:50	A1

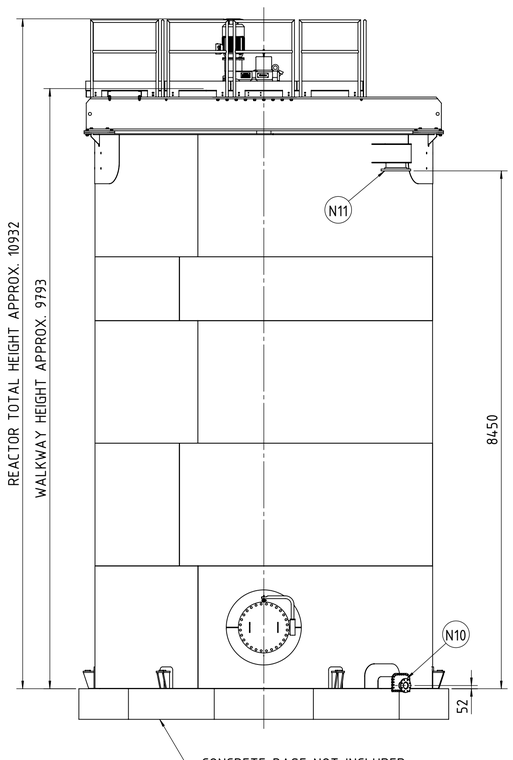
MANSOURAH-MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
PRESSURE OXIDATION
3410-REA-003 POX RECYCLE TANK

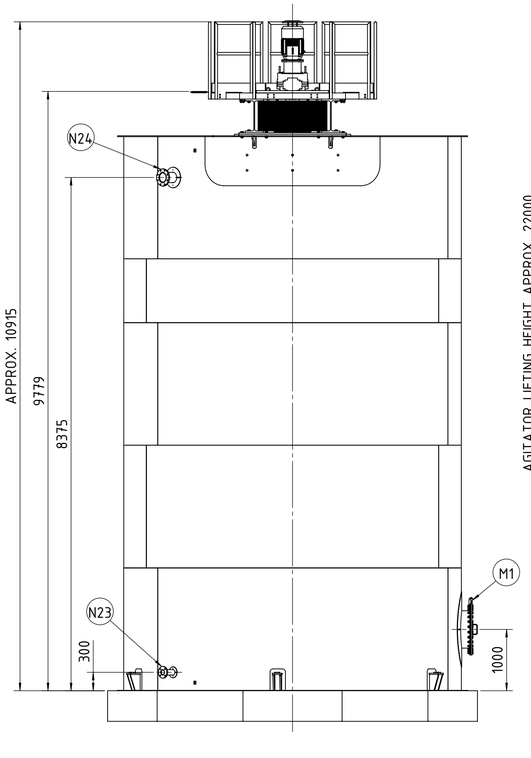
DRAWING NUMBER MD-216-3410-EG-ME-GAD-1008 REV A01

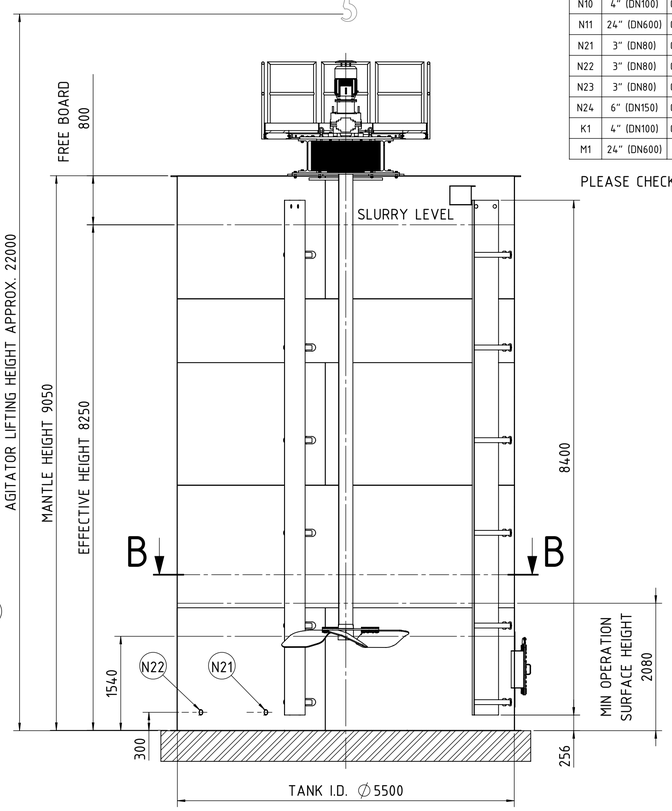
DRAWING NUMBER: MD-216-3410-EG-ME-GAD-1010
REV: A01

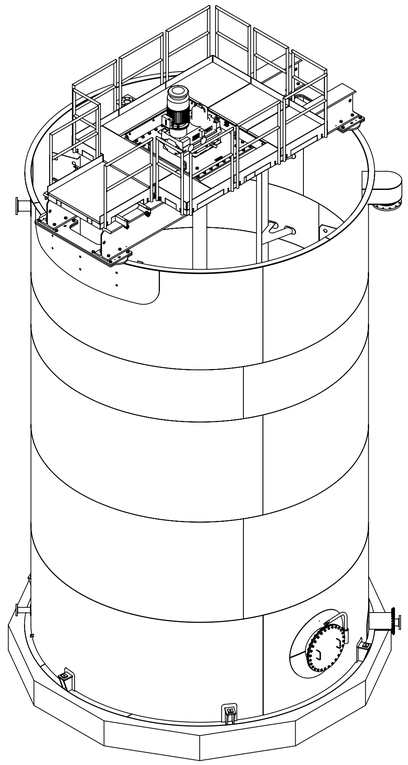
NOZZLE TABLE								
No.	SIZE	RATING	DIRECTION	HEIGHT	STANDARD	SERVICE	QUANTITY	REMARKS
N10	4" (DN100)	CLASS 150	45°	52	ASME B16.5	DRAIN	1	
N11	24" (DN600)	CLASS 150	45°	8450	ASME B16.5	OVERFLOW	1	
N21	3" (DN80)	CLASS 150	315°	300	ASME B16.5	SLURRY OUTLET	1	
N22	3" (DN80)	CLASS 150	315°	300	ASME B16.5	SLURRY OUTLET	1	
N23	3" (DN80)	CLASS 150	225°	300	ASME B16.5	DENSITY PUMP	1	
N24	6" (DN150)	CLASS 150	225°	8375	ASME B16.5	SLURRY OUTLET	1	
K1	4" (DN100)	CLASS 150	270°	9779	ASME B16.5	LEVEL MEASUREMENT	1	
M1	24" (DN600)	-	90°	1000	OUTOTEC	MANHOLE	1	

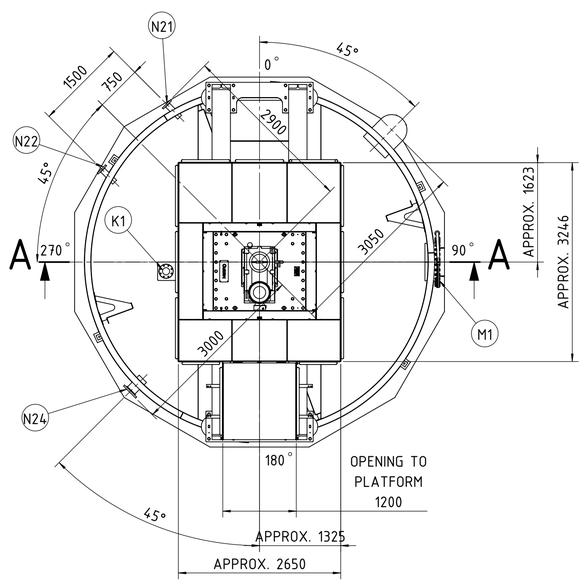
PLEASE CHECK DATASHEET MD-216-3410-EG-ME-DAT-1002 FOR PROCESS VALUES



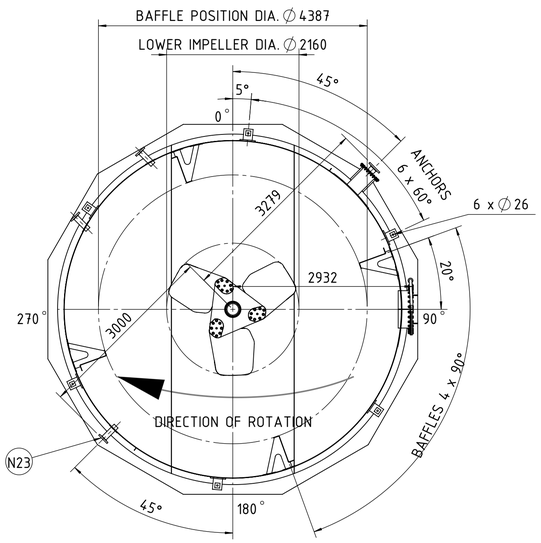








SECTION A-A



SECTION B-B

POX RECYCLE TANK	
DEAD WEIGHT (kg)	18300
OPERATIONAL WEIGHT (kg)	359900
MAXIMUM WEIGHT (kg)	393000

Document Number
MD-216-3410-EG-ME-GAD-1010

Revision A01

Approved

Mr Doron Golan - Bechtel Global

Feb 12, 2020, 6:49 PM GMT+2:00

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DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3410-EG-ME-PLN-1003	FOUNDATION AND LOAD PLAN								
MD-216-3410-EG-ME-DAT-1002	EQUIPMENT DATASHEET								
MD-216-3410-EG-PR-PID-1013	P&I DIAGRAM								
MD-216-3410-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	19.12.2019	ISSUED FOR APPROVAL	AT	TM	JR	GR	

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LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-G-02647-004 — REVISION: 0

OUTOTEC DRAWING NUMBER: 900707 GAU01 HH12 MTL03 34104 — DOCUMENT ID: OU601239486 — REVISION: 0

DRAWN	aritik	18.12.2019
CHECKED	tuoman	19.12.2019
ENGINEER	juhris	19.12.2019
APPROVED	GeoRie	19.12.2019
EMPLOYER		
SCALE	1:50	A1

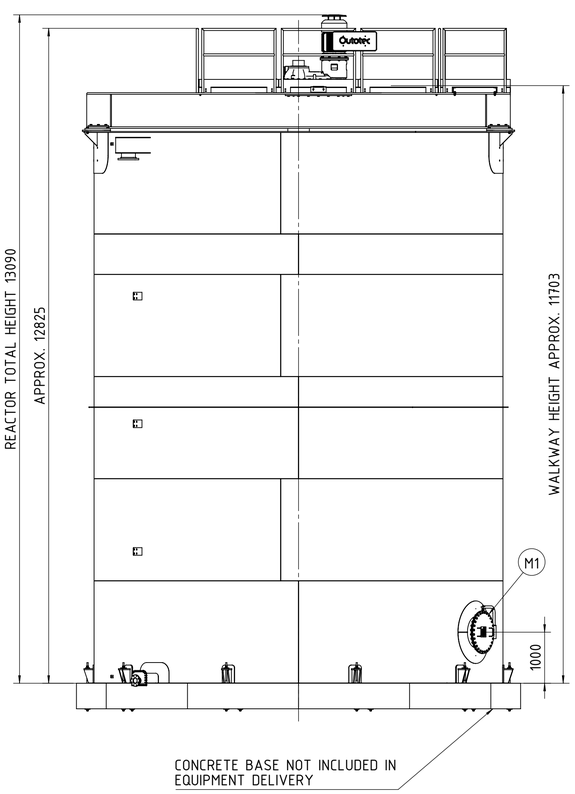
MANSOURAH-MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
PRESSURE OXIDATION
3410-REA-004 POX RECYCLE TANK

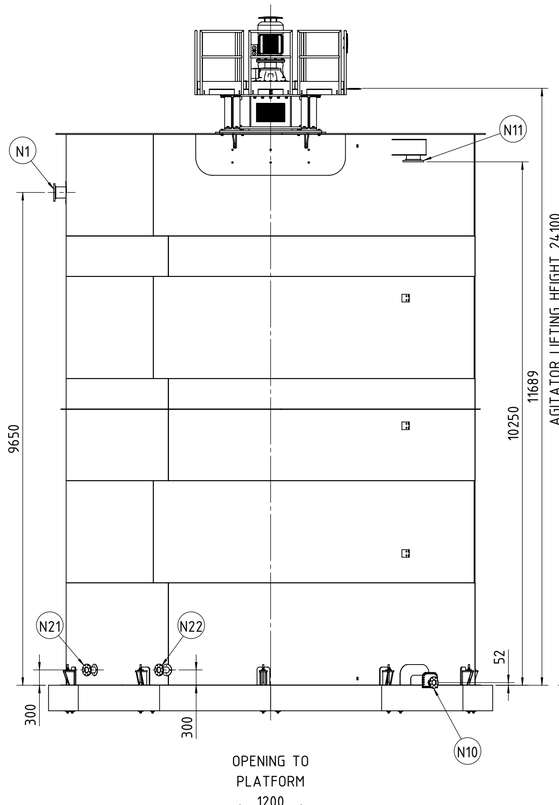
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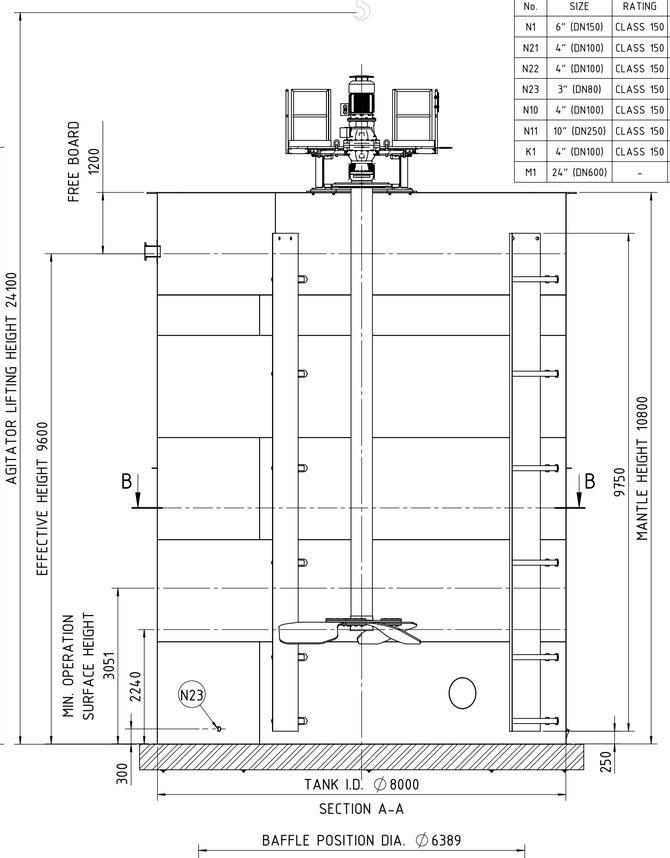
DRAWING NUMBER MD-216-3410-EG-ME-GAD-1011 | REV A01

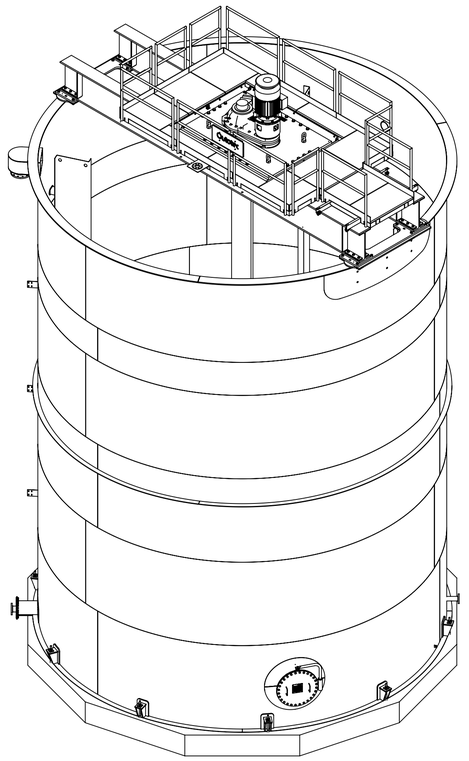
NOZZLE TABLE								
No.	SIZE	RATING	DIRECTION	HEIGHT	STANDARD	SERVICE	QUANTITY	REMARKS
N1	6" (DN150)	CLASS 150	270°	9650	ASME B16.5	SLURRY INLET	1	
N21	4" (DN100)	CLASS 150	225°	300	ASME B16.5	SLURRY OUTLET	1	
N22	4" (DN100)	CLASS 150	225°	300	ASME B16.5	SLURRY OUTLET	1	
N23	3" (DN80)	CLASS 150	315°	300	ASME B16.5	DENSITY MEASUREMENT PUMP	1	
N10	4" (DN100)	CLASS 150	135°	52	ASME B16.5	DRAIN	1	
N11	10" (DN250)	CLASS 150	140°	10250	ASME B16.5	EMERGENCY OVERFLOW	1	
K1	4" (DN100)	CLASS 150	90°	11689	ASME B16.5	LEVEL MEASUREMENT	1	
M1	24" (DN600)	-	30°	1000	OUTOTEC	MANHOLE	1	

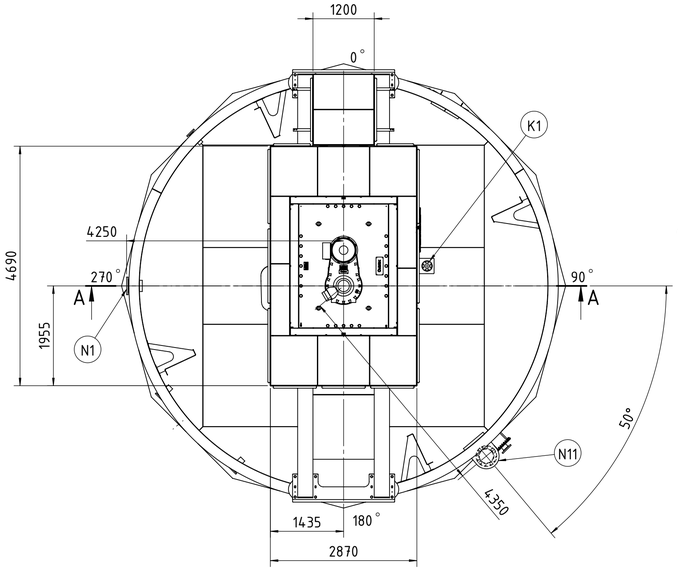
PLEASE CHECK DATASHEET MD-216-3410-EG-ME-DAT-1001 FOR PROCESS VALUES

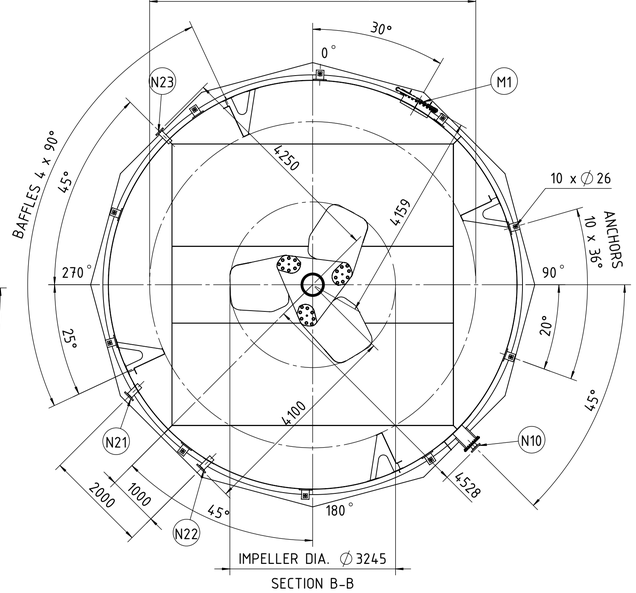












POX FEED TANK	
AGITATOR WEIGHT APPR. (kg)	6700
DEAD WEIGHT (kg)	32800
OPERATIONAL WEIGHT (kg)	748800
MAXIMUM WEIGHT (kg)	838300

BECHTEL

Document Number
MD-216-3410-EG-ME-GAD-1011

Revision A01

Approved

Mr Doron Golan - Bechtel Global

Feb 6, 2020, 3:05 PM GMT+2:00

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DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3410-EG-ME-PLN-1002	FOUNDATION AND LOAD PLAN								
MD-216-3410-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3410-EG-PR-PID-1003	P&I DIAGRAM								
MD-216-3410-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	19.12.2019	ISSUED FOR APPROVAL	VH	TM	JR	GR	

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LARSEN & TOUBRO DRAWING NUMBER 019045-450-M-G-02647-002 | REVISION 0

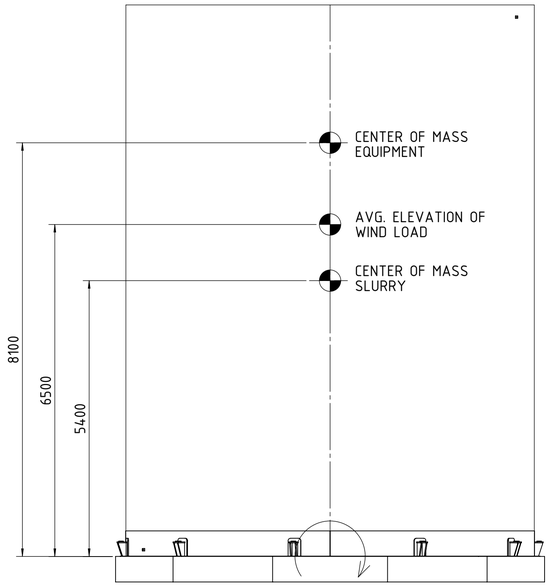
OUTOTEC DRAWING NUMBER 900707 GAU01 HH12 MTL03 34102 | DOCUMENT ID OU601239302 | REVISION 0

DRAWN	aritik	19.12.2019
CHECKED	tuoman	19.12.2019
ENGINEER	juhris	19.12.2019
APPROVED	GeoRie	19.12.2019
EMPLOYER		
SCALE	1:60	A1

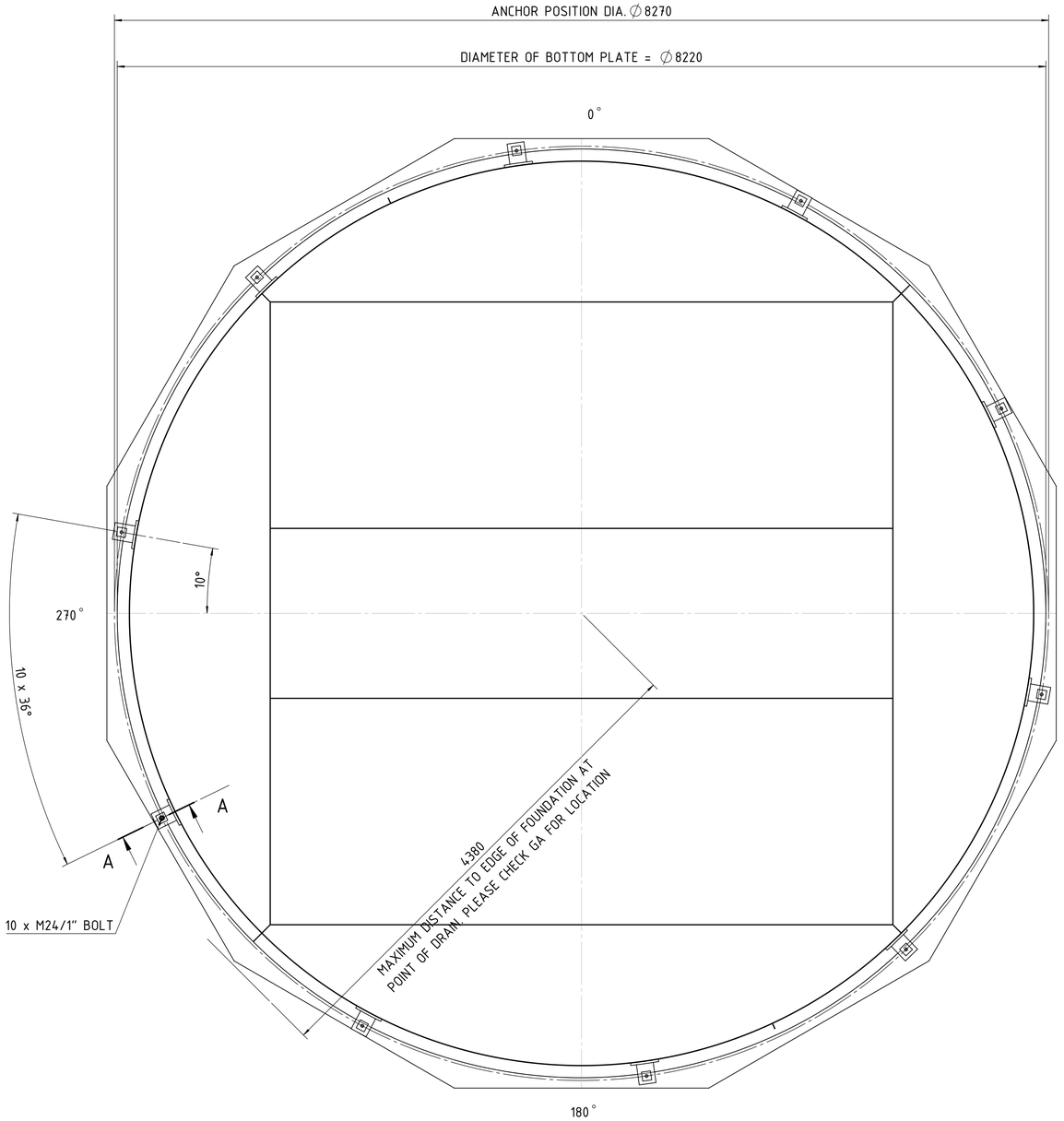
MANSOURAH-MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
PRESSURE OXIDATION
3410-REA-002 POX FEED TANK

DRAWING NUMBER MD-216-3410-EG-ME-GAD-1011 | REV A01

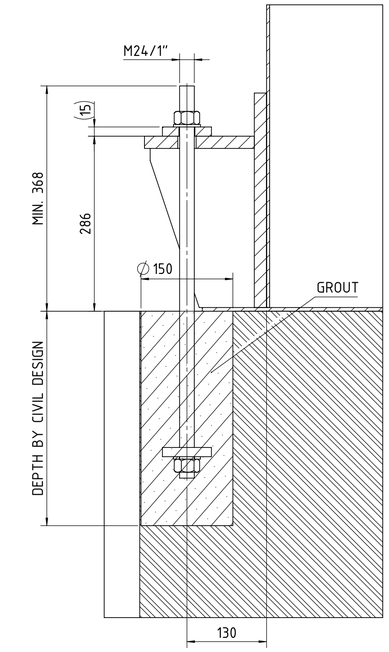
DRAWING NUMBER MD-216-3410-EG-ME-PLN-1001 REV A01



SCALE 1 : 50



LOADING NOTES:	
1	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL
CONCRETE NOTES: (WHERE CELLS ARE INSTALLED ON CONCRETE AND HOLD DOWN BOLTS ARE USED)	
1.	CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT



SECTION A-A
SCALE 1 : 5

POX FEED TANK	
DEAD WEIGHT	32750 kg
OPERATIONAL WEIGHT OF SLURRY	716000 kg
MAXIMUM WEIGHT OF SLURRY	805500 kg
EXTERNAL PLATFORM LOAD	3650 kg
NORMAL OPERATING SPEED	35,3 rpm
WIND LOAD	83 kN
SEISMIC LOAD	329 kN
SEISMIC TORQUE AT FOUNDATION LEVEL	1811 kNm

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. groud elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

BECHTEL

Document Number
MD-216-3410-EG-ME-PLN-1001

Revision A01

Approved

Mr Doron Golan - Bechtel Global

Jan 17, 2020, 10:22 PM GMT+2:00

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DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3410-EG-ME-GAD-1007	GENERAL ARRANGEMENT DRAWING								
MD-216-3410-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3410-EG-PR-PID-1002	P&I DIAGRAM								
MD-216-3410-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	19.12.2019	ISSUED FOR APPROVAL	JJ	TM	JR	GR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER 019045-450-M-D-02648-001 REVISION 1

OUTOTEC DRAWING NUMBER 900707 GAU01 HH12 MLD15 34101 DOCUMENT ID OU601263694 REVISION 1

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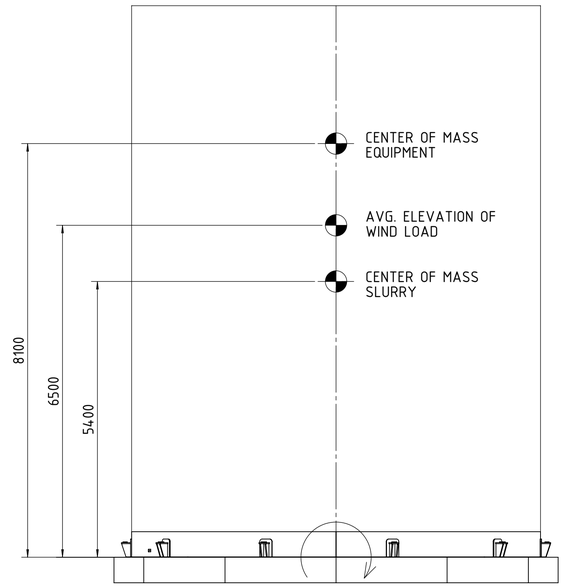
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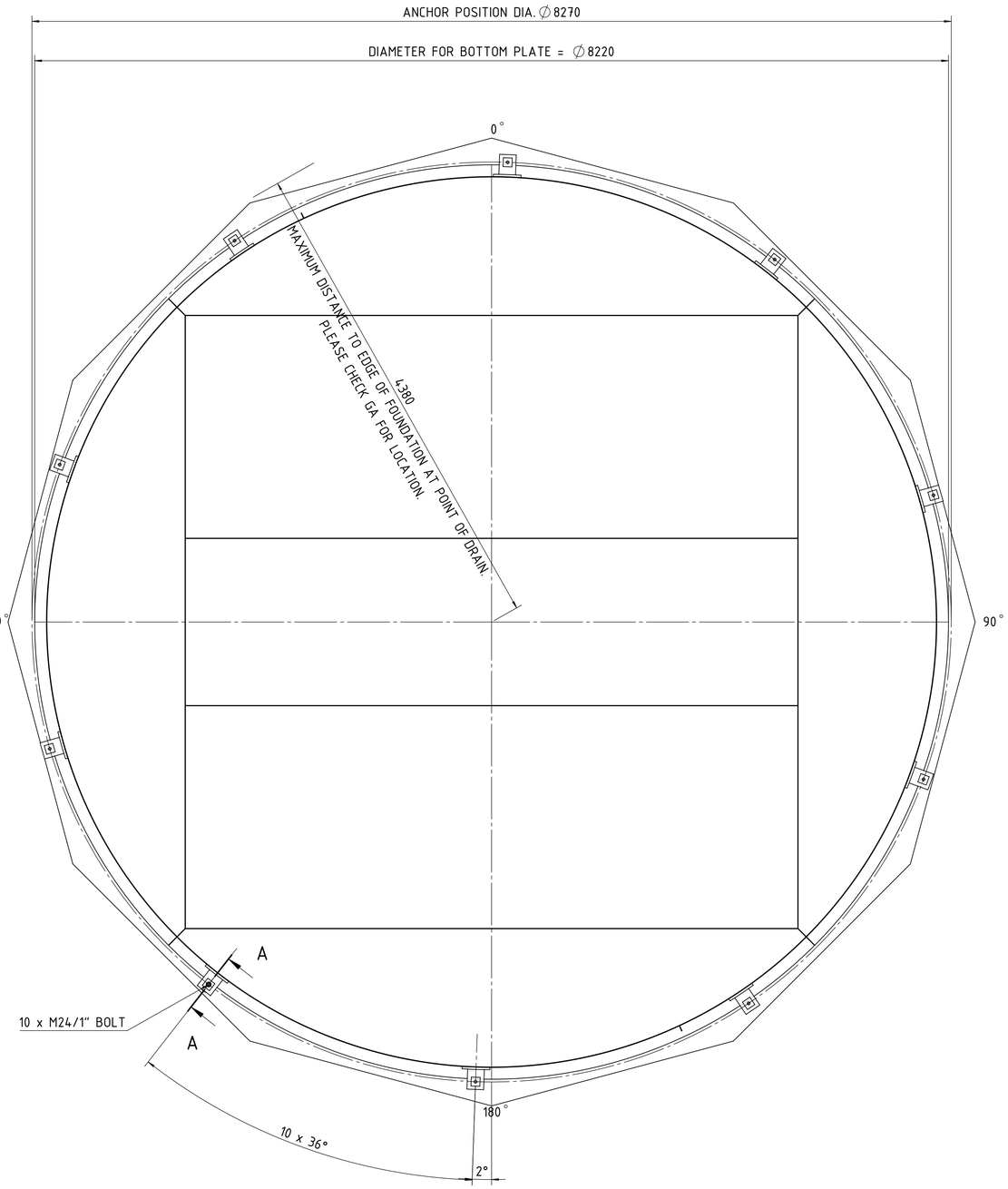
DRAWN	joujaa	16.12.2019	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	tuoman	19.12.2019	FOUNDATION AND LOAD PLAN
ENGINEER	juhris	19.12.2019	PRESSURE OXIDATION
APPROVED	GeoRie	19.12.2019	3410-REA-001 POX FEED TANK
EMPLOYER			
SCALE	1:25	A1	DRAWING NUMBER MD-216-3410-EG-ME-PLN-1001 REV A01

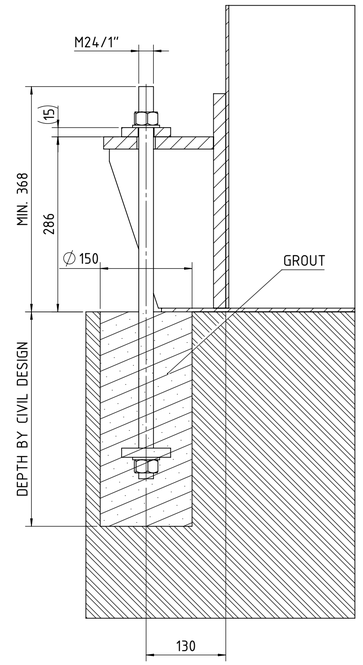
DRAWING NUMBER	REV
MD-216-3410-EG-ME-PLN-1002	A01

LOADING NOTES:	
1	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL
CONCRETE NOTES: (WHERE CELLS ARE INSTALLED ON CONCRETE AND HOLD DOWN BOLTS ARE USED)	
1.	CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT



SCALE 1 : 50





SECTION A-A
SCALE 1 : 5

POX FEED TANK	
DEAD WEIGHT	32750 kg
OPERATIONAL WEIGHT OF SLURRY	716000 kg
MAXIMUM WEIGHT OF SLURRY	805500 kg
EXTERNAL PLATFORM LOAD	3650 kg
NORMAL OPERATING SPEED	35,3 rpm
WIND LOAD	83 kN
SEISMIC LOAD	329 kN
SEISMIC TORQUE AT FOUNDATION LEVEL	1811 kNm

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. groud elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

BECHTEL

Document Number
MD-216-3410-EG-ME-PLN-1002

Revision A01

Approved

Mr Doron Golan - Bechtel Global

Jan 17, 2020, 10:23 PM GMT+2:00

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DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3410-EG-ME-GAD-1008	GENERAL ARRANGEMENT DRAWING								
MD-216-3410-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3410-EG-PR-PID-1003	P&I DIAGRAM								
MD-216-3410-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	19.12.2019	ISSUED FOR APPROVAL	JJ	TM	JR	GR	

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LARSEN & TOUBRO DRAWING NUMBER	REVISION
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OUTOTEC DRAWING NUMBER: 900707 GAU01 HH12 MLD15 34102	DOCUMENT ID: OU601264246 / REVISION: 1

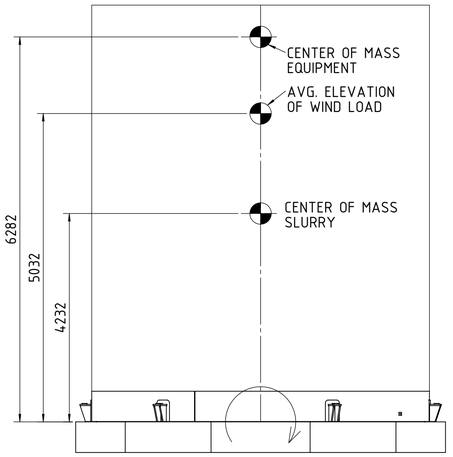
DRAWN	joujaa	16.12.2019
CHECKED	tuoman	19.12.2019
ENGINEER	juhris	19.12.2019
APPROVED	GeoRie	19.12.2019
EMPLOYER		
SCALE	1:25	A1

MANSOURAH-MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
PRESSURE OXIDATION
3410-REA-002 POX FEED TANK

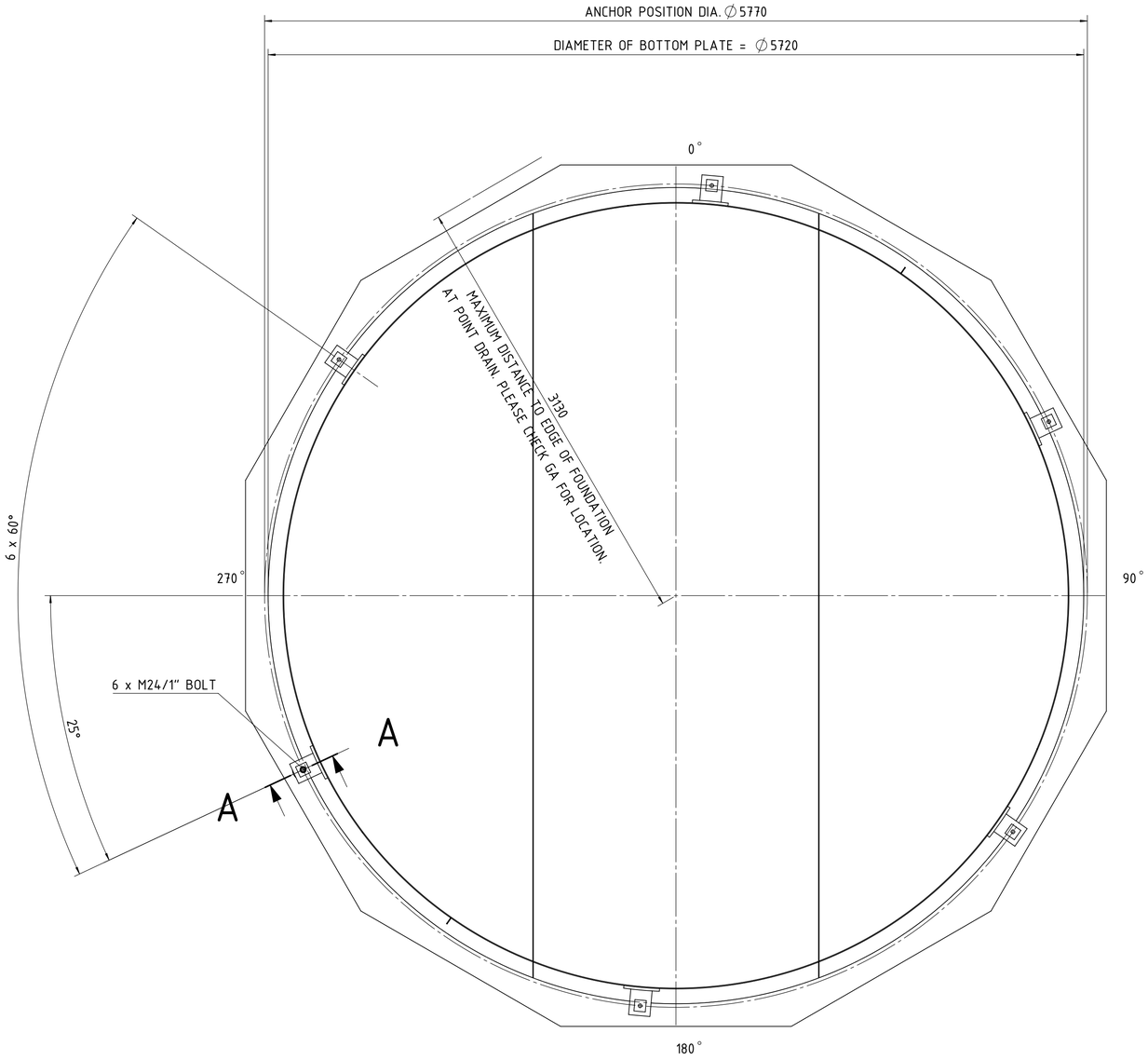
DRAWING NUMBER	REV
MD-216-3410-EG-ME-PLN-1002	A01

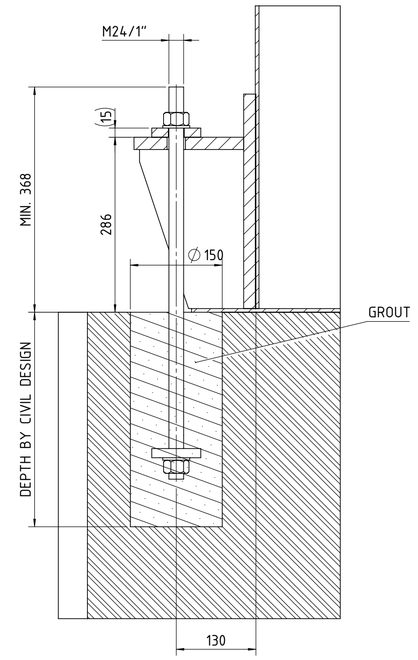
DRAWING NUMBER MD-216-3410-EG-ME-PLN-1003 REV A01

LOADING NOTES:	
1	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL
CONCRETE NOTES: (WHERE CELLS ARE INSTALLED ON CONCRETE AND HOLD DOWN BOLTS ARE USED)	
1.	CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT



SCALE 1 : 50





SECTION A-A
SCALE 1 : 5

POX RECYCLE TANK	
DEAD WEIGHT	18300 kg
OPERATIONAL WEIGHT OF SLURRY	341700 kg
MAXIMUM WEIGHT OF SLURRY	374800 kg
EXTERNAL PLATFORM LOAD	2400 kg
NORMAL OPERATING SPEED	50,5 rpm
WIND LOAD	48 kN
SEISMIC TORQUE AT FOUNDATION LEVEL	668 kNm

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. groud elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

BECHTEL

Document Number
MD-216-3410-EG-ME-PLN-1003

Revision A01

Approved

Mr Doron Golan - Bechtel Global

Jan 17, 2020, 10:23 PM GMT+2:00

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DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3410-EG-ME-GAD-1008,1010	GENERAL ARRANGEMENT DRAWING								
MD-216-3410-EG-ME-DAT-1002	EQUIPMENT DATASHEET								
MD-216-3410-EG-PR-PID-1012,1013	P&I DIAGRAM								
MD-216-3410-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	19.12.2019	ISSUED FOR APPROVAL	JJ	TM	JR	GR	

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MANSOURAH - MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
PRESSURE OXIDATION
3410-REA-003, 004 POX RECYCLE TANK

DRAWING NUMBER MD-216-3410-EG-ME-PLN-1003 REV A01